

and artificial intelligence.

Utilising an Arduino Uno micro-controller, it offers a user-friendly interface enabling remote control via Bluetooth connectivity from a smartphone or other compatible devices. Additionally, the integration of voice control enhances accessibility and convenience in operating the robot. Fig. 1 depicts the author's prototype, and the components used to build this robot are listed in the Bill of Materials table.

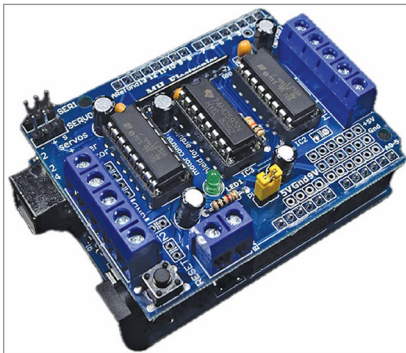


Fig. 3: Motor driver shield with Arduino Uno

The robot is built around Arduino Uno (MOD1), motor driver shield L293D (MOD2), ultrasonic sensor HC-SR04 (S1), Bluetooth module HC-05, four geared DC motors (M1 through M4), and a servo motor, with additional components such as robot chassis and jumper wires used for assembly. Fig. 2 illustrates the circuit diagram of the motor driver shield with Arduino Uno, while Fig. 3 shows the motor driver shield with Arduino Uno board.

The motor driver shield (MOD2) does not require wiring with the Arduino Uno; it just fits over the Arduino headers.

Construction and testing

Install the 'AFMOTOR' library in Arduino IDE, select the board as Arduino Uno, and upload the source code. Fig. 5 shows a snippet of the source code.

Upload the source code into the Arduino Uno and interconnect

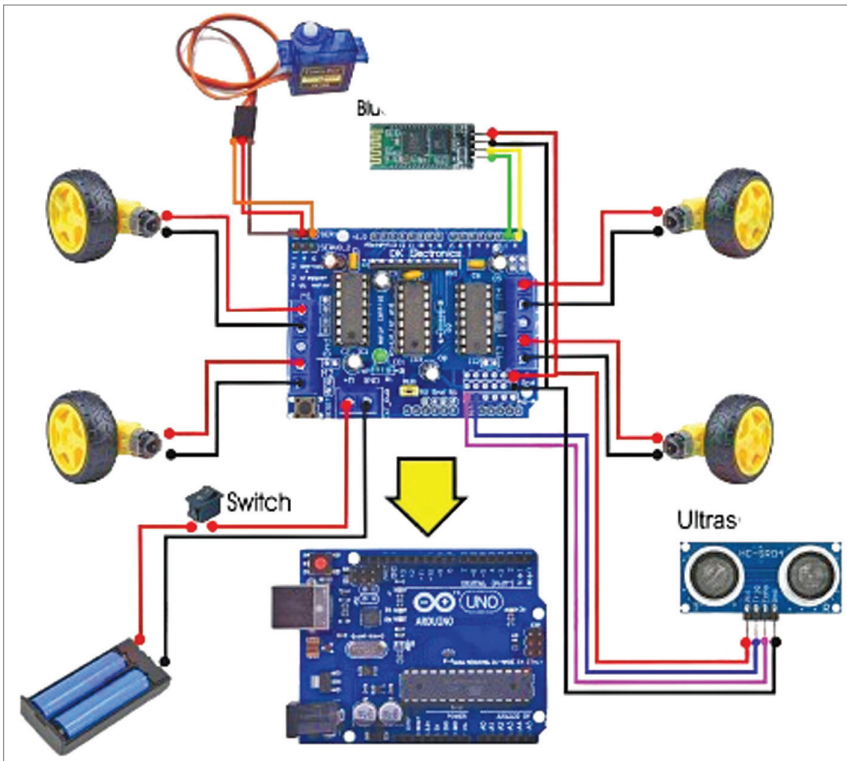


Fig. 4: Connections diagram

the motor driver shield with it, as shown in Fig. 3. Then, complete the connections of other components with the terminals of the motor driver shield, as shown in Fig. 4. The hardware is now ready to use.

Power on the robot and connect the robot's Bluetooth with any Bluetooth Arduino voice control app on your Android phone. Then, send the command to move forward or backward; the robot will move accordingly. Refer to the link:

sketch_apr17a.ino

```

1
2 #include <Servo.h>
3 #include <AFMotor.h>
4
5 #define Echo A0
6 #define Trig A1
7 #define motor 10
8 #define Speed 170
9 #define spoint 103
10
11 char value;
12 int distance;
13 int Left;
14 int Right;
15 int L = 0;
16 int R = 0;
17 int L1 = 0;
18 int R1 = 0;
19
20 Servo servo;
21 AF_DCMotor M1(1);
22 AF_DCMotor M2(2);
23 AF_DCMotor M3(3);
24 AF_DCMotor M4(4);
25
26 void setup() {
27   Serial.begin(9600);
28   pinMode(Trig, OUTPUT);
29   pinMode(Echo, INPUT);
30   servo.attach(motor);
31   M1.setSpeed(Speed);
32   M2.setSpeed(Speed);
33   M3.setSpeed(Speed);
34   M4.setSpeed(Speed);
35 }
36 void loop() {
37   //Obstacle();
38   //Bluetoothcontrol();
39   voicecontrol();
40 }
41
42 void Bluetoothcontrol() {

```

Fig. 5: Snippet of the source code